

Sarvesh Bhattacharyya

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PROFILE SUMMARY

Final-year ECE student at MSRIT and AI Engineer with production experience in multi-agent systems, RAG and voice AI. Focused on systems that work under real constraints: latency, cost, deployment, and reliability.

EXPERIENCE

AI Engineer Intern — Beaut Group (business.superowl.in)

Bengaluru, Karnataka · On-site (Feb 2026 – Present)

- Independently designed and shipped a near-production multi-tenant AI automation platform for small businesses as a solo intern — given only a problem statement, no codebase, no supervision. Built end-to-end from infrastructure to product, handling real customer interactions across multiple live business clients simultaneously. Reduced end-to-end response latency from ~10s to ~3s.
- Engineered real-time communication and control workflows enabling business owners to stay aware of and intervene in live customer interactions instantly, without manual effort — increasing owner responsiveness while keeping operations fully automated by default.
- Delivered the complete product solo across backend, integrations, and a React Native mobile app with real-time notifications — taking a B2B SaaS product from blank problem statement to near-production across the full stack in under 3 months.

PROJECTS

1. ROAST — Live-Market Resume Intelligence System

 [GitHub](#)

- ROAST pulls live market intelligence for your exact role, city, and company type before reading a single line — feedback calibrated against your actual competition this week. Consists of a 6-agent pipeline including a TechnicalDepthAgent that web-searches via DuckDuckGo MCP server for unfamiliar technologies in real time before evaluating them, so it understands what you built rather than keyword-matching. Optional JD input parsed into structured requirements fed to all agents.
- Built the market intelligence store entirely offline — monthly ingestion pipeline scrapes Naukri, Reddit, Levels.fyi, Wellfound, and TeamBlind into SQLite+FTS5+sqlite-vec.
- Reduced ingestion time from 10 hours to 1 hour by verifying actual API rate limits and tuning semaphore concurrency accordingly. DIVE retrieval at query time: 6 targeted queries → parallel BM25 + vector → RRF fusion → Groq distillation. Results stream section-by-section via WebSocket, Redis-backed session state survives disconnects and container restarts.
- ReviewAgent falls back across 5 LLM providers automatically with circuit breakers preventing cascade failures; proactive provider switching triggers before rate limits hit. Quality gate rejects outputs below 400 words or missing follow-up questions.

Skills / Tools: FastAPI · asyncio · Redis · WebSocket · Groq · Langfuse · instructor · SQLite + FTS5 · React+Vite

2. ACARE — Autonomous Clinical Assistance Robot | Institutionally Funded Final year Team Project

- Architected and developed the complete high-level intelligence stack for a 6-DOF clinical Robot on ROS2/Raspberry Pi 5 — 10+ nodes with typed topics, custom QoS policies, hierarchical state machine (10 states), and full integration boundary (CAN/UART heartbeat + command protocol).
- Engineered a custom YOLOv11 perception engine by self-curating and annotating a proprietary medical tool dataset from scratch; achieved 97.5% mAP for high-precision surgical instrument tracking.
- Built a real-time voice & safety pipeline integrating Deepgram/Groq/LangGraph with a Bayesian Next-Best-View search strategy for reduced search time and a graded safety system (WARNING/CRITICAL/ESTOP) for dynamic autonomous handovers.

Skills / Tools: ROS2 · Python · C++ · YOLOv11 TFLite INT8 · LangGraph · Groq · Deepgram · MediaPipe · SQLite

SKILLS

Languages: Python, JavaScript, C++, Rust(basic)

AI/ML: Multi-Agent Systems, RAG Pipelines, LLM Fine-tuning (QLoRA), Voice AI, LangChain, LangGraph, Langfuse, Prompt engineering

Infra & Data: Docker, GitHub Actions, Redis, SQLite, Neo4j, Vector DBs (Qdrant, ChromaDB, Pinecone), Firebase

Frameworks & Tools: FastAPI, React Native, GitHub, Bash, Ubuntu

EDUCATION

MS Ramaiah Institute of Technology, Bengaluru (2022 – 2026)

B.E. Electronics & Communication Engineering | CGPA: 7.47 / 10

Relevant Coursework: Data Structures, Operating Systems, Computer Networks, Python, C/C++

Machine Learning Specialization — DeepLearning.AI / Andrew Ng, Coursera | 2026